AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 32 · WORKBOOK

Build a Paper Prototype

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected prototype handover and run record.

Make the materials, collect a partner's actual trace, diagnose an instruction defect and build the independent equipment prototype.

Keep one habit: Keep requests, catalogue, instruction versions, expected results and observed traces together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Design the Decision **Process**. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Find a Real Problem. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Prompts Are Design **Instruction**s. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Design the Decision Process

Find a Real Problem

Prompts Are Design Instructions

W1 · Make complete prototype materials

> Use the supplied standalone catalogue and instruction-card reference.

Your evidence record

- Standalone activity materials: catalogue B version 1 is B1 Bookmark drawing (10 min, quiet Yes), B2 Story circle (8 min, quiet No), B3 Pattern puzzle (12 min, quiet Yes), B4 Tower challenge (15 min, quiet No), in that order. Valid whole minutes are 0–30 and Quiet required is Yes/No. A fit requires time≤available minutes AND (quiet not required OR activity quiet). P1 version 1: Card 1 validates both fields, clarifies invalid input and stops, otherwise starts marker at B1 with cleared output. Card 2 checks both conditions, records them and stops with the first fitting row and reason. Card 3 advances one row after failure and returns to Card 2, or stops with no match after the last row fails. Start every new request at Card 1. Save request, catalogue/instruction versions, expected result and actual trace separately.

- Create P1 version 1 using the three **instruction** cards in the lesson. Copy catalogue B version 1 and make a request card, current-row marker and **trace** sheet. List what each item records.

  Explain which information stays fixed during one run and what changes.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Predict, hand over and record the run

> Keep the expected result separate from actual operator steps.

Your evidence record

- Use the manual checklist to predict P1's **output** for 11 minutes No. Then ask a partner to operate your saved cards. Record the **expected result** separately from the observed steps and **output**, including the reason.

  Explain why the **operator** should stop without checking later rows.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Diagnose and version an instruction defect

> Preserve the original failure and label the changed instructions.

Your evidence record

- Saved **instruction**s P0 say "start at B1; if B1 fails, report no match." For the valid request 9 minutes No, an **operator** follows P0 exactly and reports no match. Identify the **instruction** defect and write the changed step.

  Explain how to label the correction and what to retain from the failed run.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Save enough evidence to inspect the result

> Retain requests, versions, expected outputs and actual traces.

Your evidence record

- A teammate keeps only a final “B2” answer and throws away the request and old **instruction**s after editing them. Explain why another person cannot check this result.

  List what to save for each run and explain why an observed error must remain in the record.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Build a new route through a small catalogue

> Keep one first attempt before hints or feedback.

Read and explain independently

Independently make and **trace instruction** cards for this paper equipment catalogue: E1 first, capacity 3 people; E2 second, capacity 6. Valid **input** is a whole group size from 1 to 8.

Suggest the first item whose capacity is at least the group size; no booking occurs. Include invalid **input** and no match. Write **expected result**s, then operate your cards for groups of 5 and 7.

Record both runs and explain every stopping point.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Build a new route through a small catalogue

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MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Build the complete paper sorter: **input** cards, **process** guide, **output** cards, not-sure card, and test log.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Run and record the first match for Building and tracing a paper **prototype**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Building and tracing a paper **prototype**. Proposed explanation: “A correct-looking answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Assign three people as **input** reader, rule follower and **output** recorder. Run the repaired sorter on paper, metal and an unreadable description. Record expected and observed outcomes. What does this **prototype** demonstrate?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A paper **prototype** has **input** cards, decision cards and paper, plastic and review trays.

How can a learner simulate it?

Show your **trace**, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Your paper **prototype** shows an **input** card, a decision card and an **output** card. Explain what a person must simulate and what the **prototype** cannot prove.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 32 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Assign three people as **input** reader, rule follower and **output** recorder. Run the repaired sorter on paper, metal and an unreadable description. Record expected and observed outcomes. What does this **prototype** demonstrate?

> **My explanation after practice**

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Independently make and **trace instruction** cards for this paper equipment catalogue: E1 first, capacity 3 people; E2 second, capacity 6. Valid **input** is a whole group size from 1 to 8.

Suggest the first item whose capacity is at least the group size; no booking occurs. Include invalid **input** and no match. Write **expected result**s, then operate your cards for groups of 5 and 7.

Record both runs and explain every stopping point.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

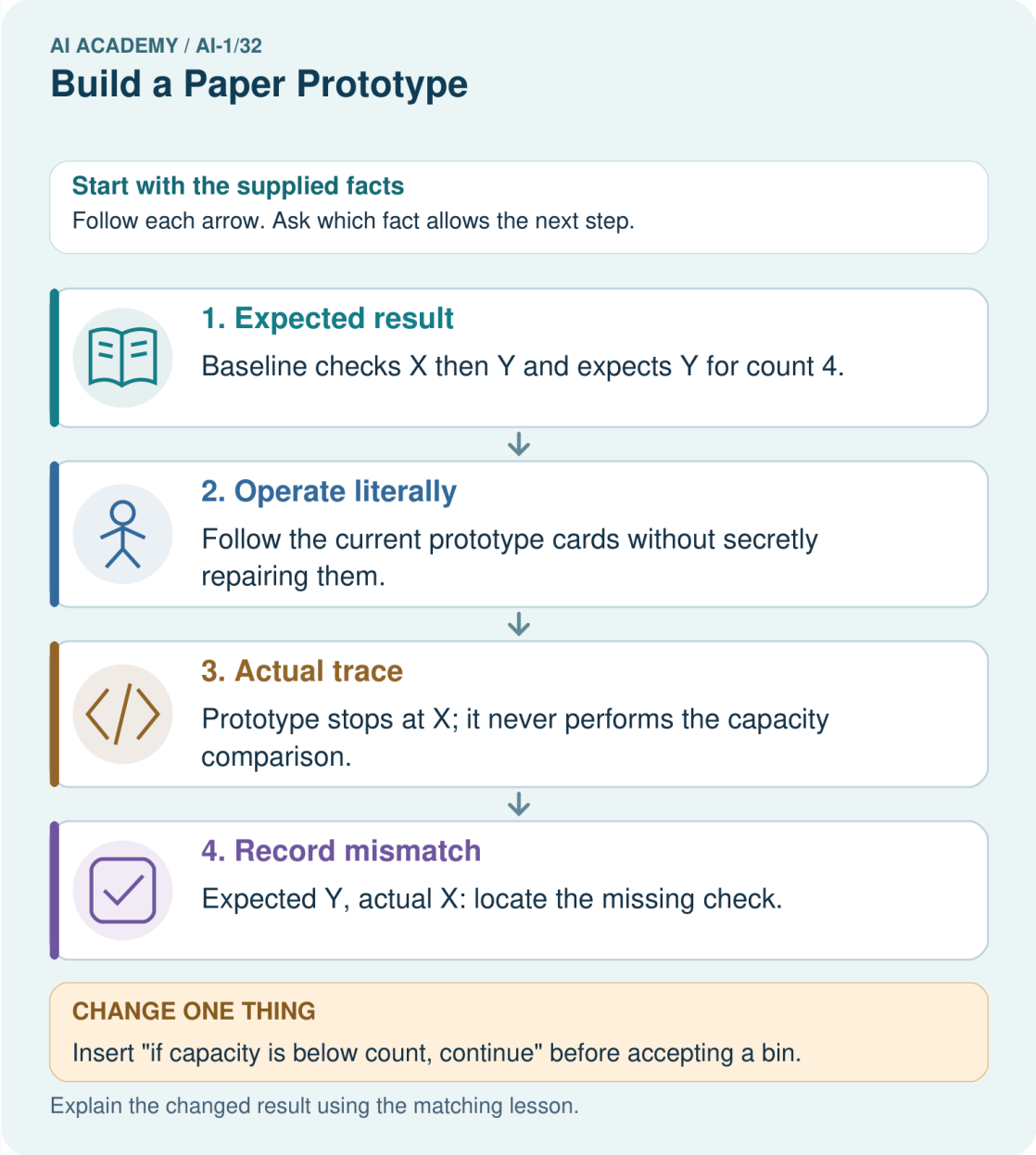
> Keep the explanation and the support you used.

Use this when helpful

A fictional team turns a storage-bin specification into movable **instruction** cards.

Bins X capacity 3 then Y capacity 7; valid count 1–9. Select first capacity at least count.

For request 4, write baseline Y before operating cards. **Prototype** cards mistakenly stop with X immediately.



Follow the labelled picture with the matching explanation. Open the vector diagram

What should appear in the test log before and after the repair?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Save sorter **instruction** cards, dated versions and **operator trace**s alongside the manual expected **output**s.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Save sorter **instruction** cards, dated versions and **operator trace**s alongside the manual expected **output**s.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional team turns a storage-bin specification into movable **instruction** cards.
- Which labelled step matches this detail? Expected Y, actual X: locate the missing check.
- 1. Actual **trace**
- 2. Record mismatch
- 3. Operate literally

- 4. **Expected result**

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Record mismatch Expected Y, actual X: locate the missing check.

A paper **prototype** is executable when another person can follow its **instruction**s without guessing your intention. Record the actual wrong result separately from the expected one.

Discovering a missing comparison is useful **evidence** for repair; pretending the **operator** followed the intended rule would hide the defect.

Changed-case question

- Changed-case practice: Insert “if capacity is below count, continue” before accepting a bin. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. The revised **trace** rejects X and accepts Y for 4.

- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial **trace**. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The revised **trace** rejects X and accepts Y for 4. 3<4 is true; 7<4 is false, so Y fits.

Repair and save your work

Review Design the Decision **Process** (ai-1/31). Goal: Design a decision **process** that covers valid, invalid and no-match requests.

Short practice: Where does the first original rejection occur, and why is K3 not originally selected?. Return to this week and explain what you changed.

Review Find a Real Problem (ai-1/29). Goal: Describe a specific user difficulty using supplied **evidence** without inventing its cause.

Short practice: Write a one-sentence project goal and say what **evidence** would count as a failure.. Return to this week and explain what you changed.

Review Prompts Are Design **Instruction**s (ai-1/26). Goal: Write a prompt with a clear task, supplied **evidence** and checkable **output** requirements.

Short practice: What must happen if B instead conflicts with another equally authoritative duration record?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Build a Paper Prototype: the case

AI ACADEMY / AI-1/32 · 01 · LOOK AT THE FACTS

Build a Paper Prototype: the case

The situation

A fictional team turns a storage-bin specification into movable instruction cards.

Supplied evidence

Bins X capacity 3 then Y capacity 7; valid count 1–9. Select first capacity at least count.
For request 4, write baseline Y before operating cards. Prototype cards mistakenly stop with X immediately.

By the end, I can build and operate a paper prototype while recording actual traces.

Follow the labelled picture with the matching explanation. Open the vector diagram

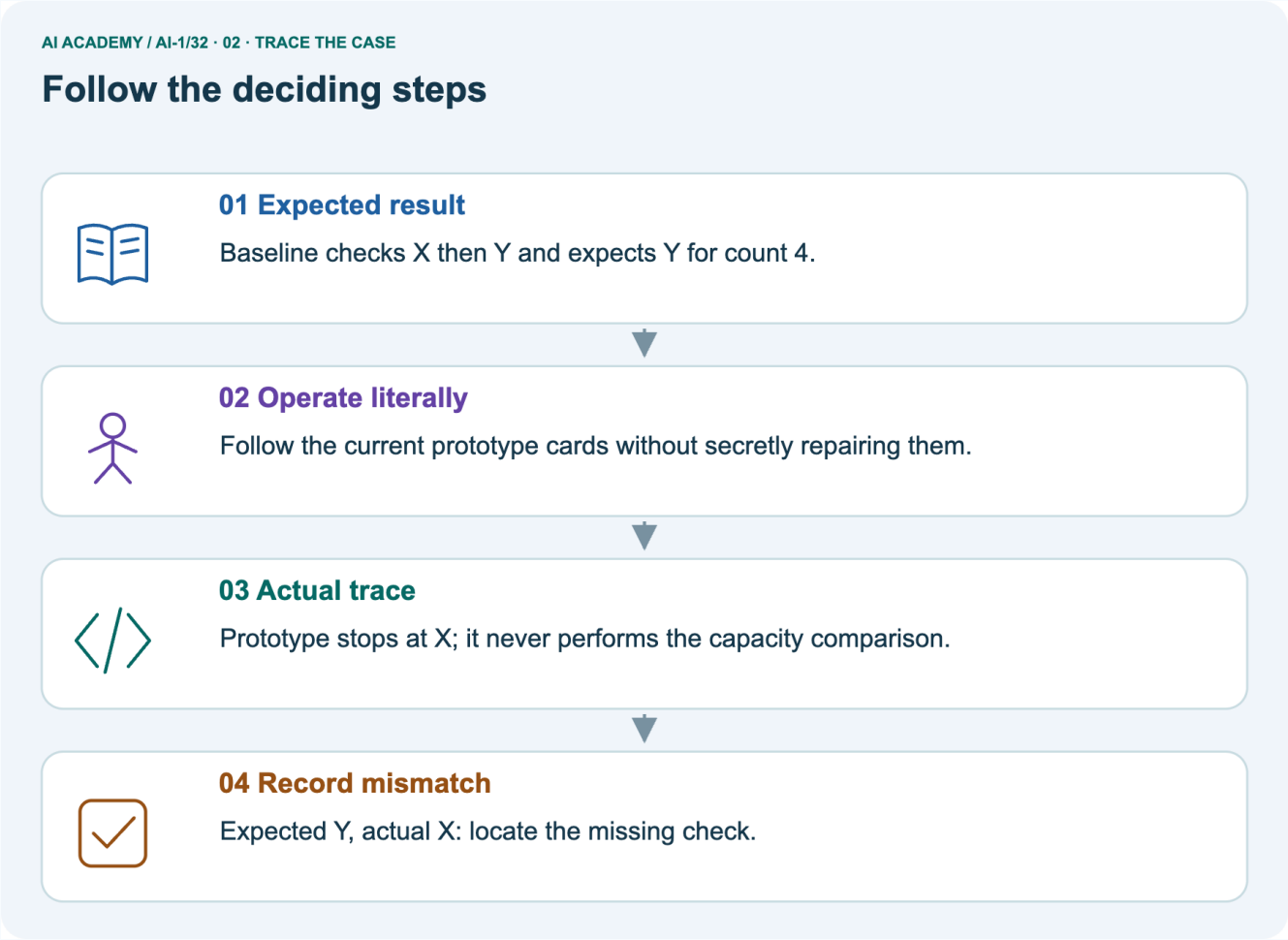
A fictional team turns a storage-bin specification into movable **instruction** cards. Supplied facts: Bins X capacity 3 then Y capacity 7; valid count 1–9. Select first capacity at least count.

For request 4, write baseline Y before operating cards. **Prototype** cards mistakenly stop with X immediately.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



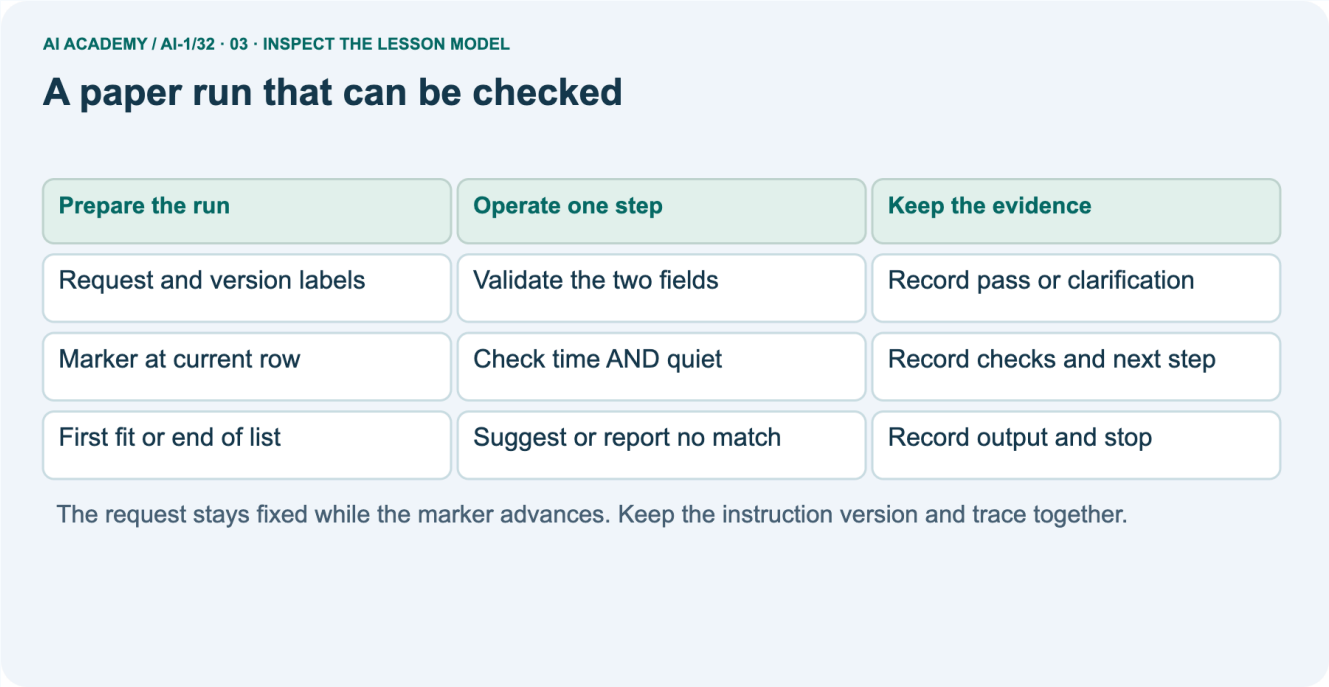
Follow the labelled picture with the matching explanation. Open the vector diagram

Expected result: Baseline checks X then Y and expects Y for count 4. → Operate literally: Follow the current **prototype** cards without secretly repairing them. → Actual **trace**: **Prototype** stops at X; it never performs the capacity comparison. → Record mismatch: Expected Y, actual X: locate the missing check.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. A paper run that can be checked



Follow the labelled picture with the matching explanation. Open the vector diagram

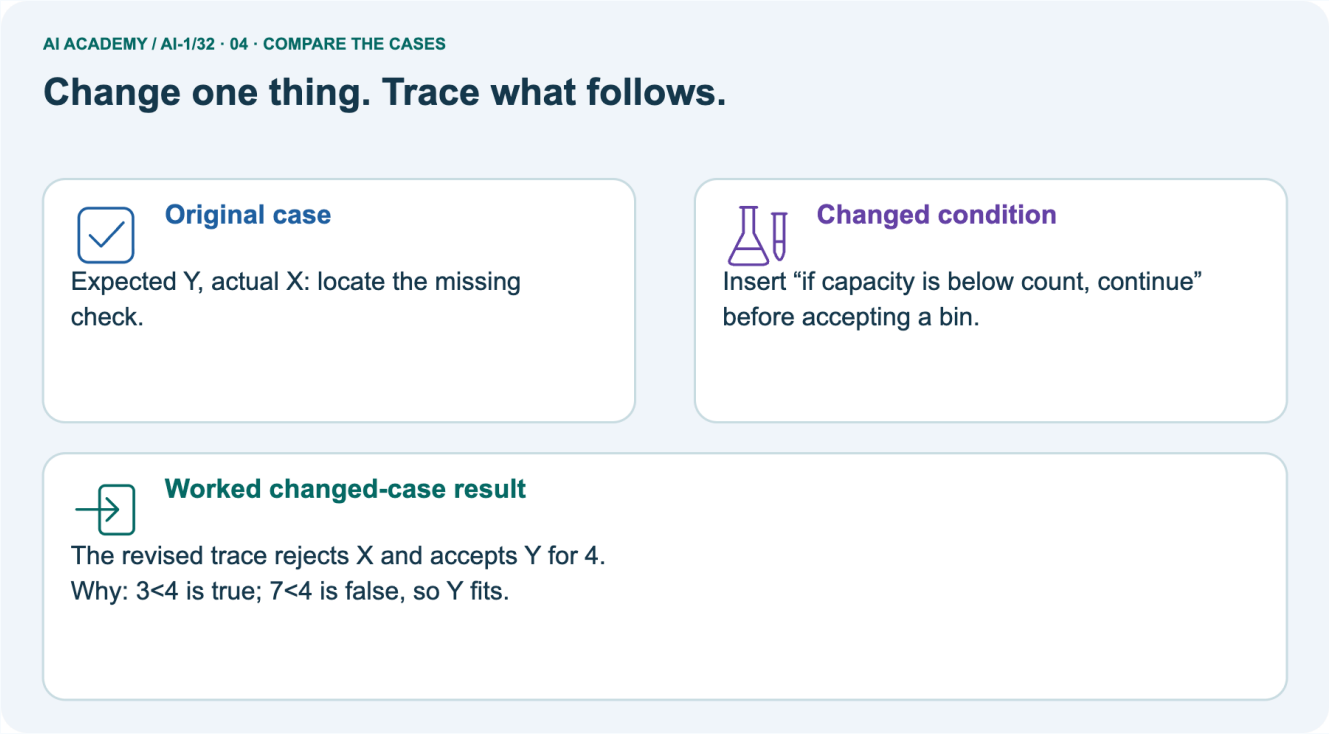
A paper run that can be checked. Request and version labels | Validate the two fields | Record pass or clarification. Marker at current row | Check time AND quiet | Record checks and next step.

First fit or end of list | Suggest or report no match | Record **output** and stop

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

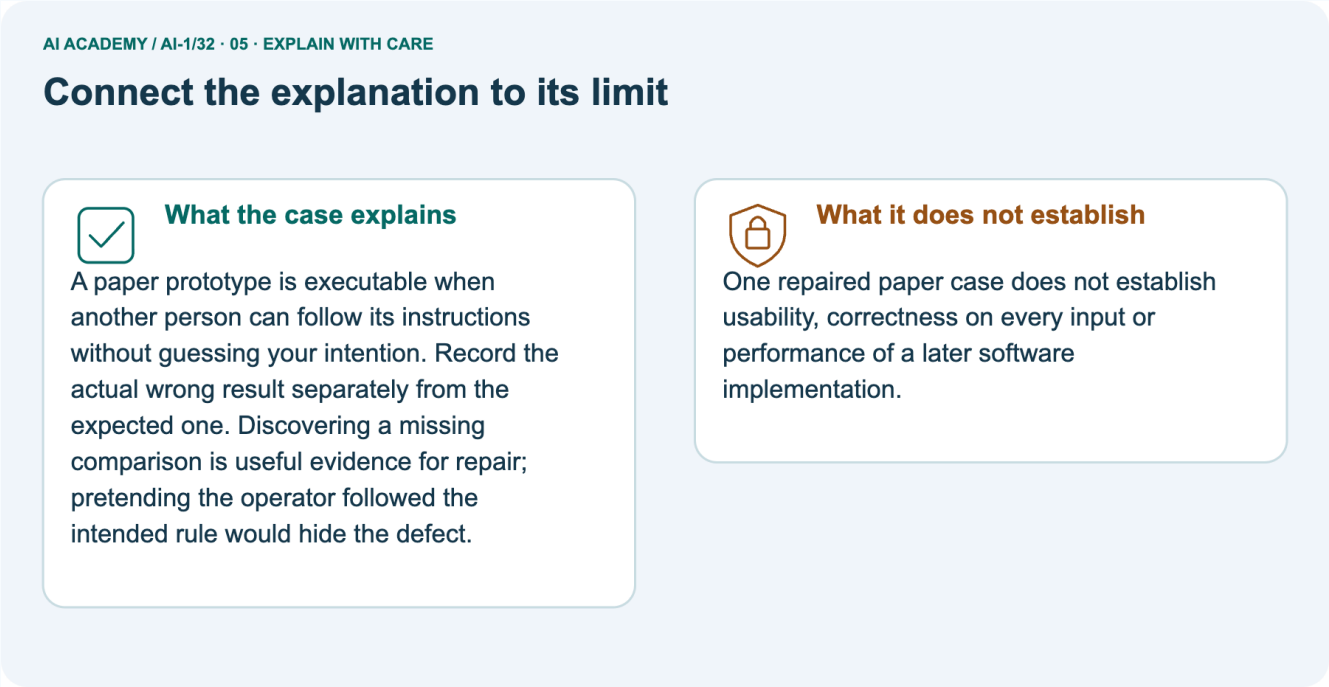
Original case: Expected Y, actual X: locate the missing check. Changed condition: Insert “if capacity is below count, continue” before accepting a bin. **Expected result**: The revised **trace** rejects X and accepts Y for 4.

Reason: 3<4 is true; 7<4 is false, so Y fits.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A paper **prototype** is executable when another person can follow its **instruction**s without guessing your intention. Record the actual wrong result separately from the expected one.

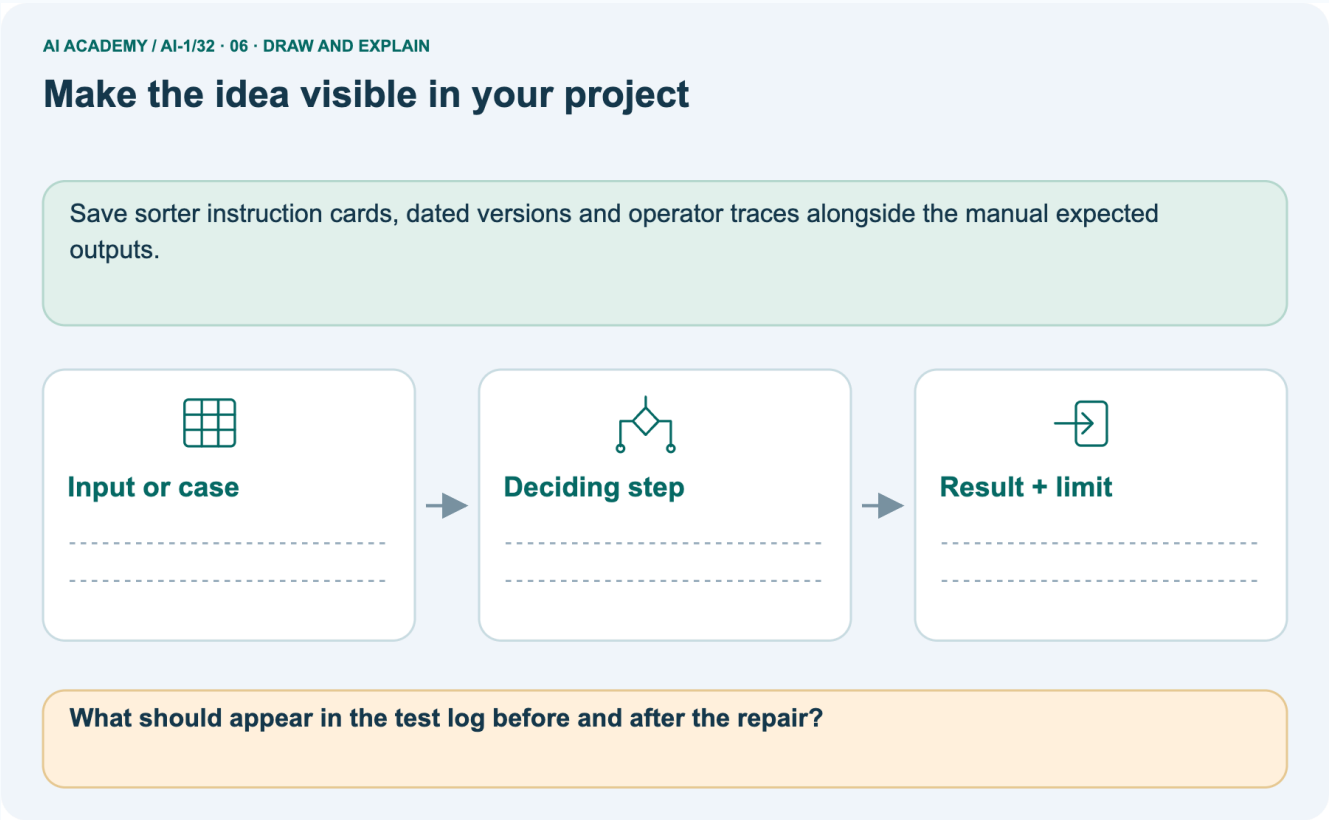
Discovering a missing comparison is useful **evidence** for repair; pretending the **operator** followed the intended rule would hide the defect.

Limit: One repaired paper case does not establish usability, correctness on every **input** or performance of a later software implementation.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Save sorter **instruction** cards, dated versions and **operator trace**s alongside the manual expected **output**s. Draw your **input**, deciding step, result and limit.

Supported question: What should appear in the test log before and after the repair?

Draw your project **trace** and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Build a Paper Prototype: the case

02. Follow the deciding steps

03. A paper run that can be checked

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.